PHYS401: Quantum Mechanics I
Homework #11

Evan Shipley-Friedt

December 10th, 2025

Problem 1: Hydrogen atom - Probability for $r < a_0$

Solution

For a separated eigenstate

$$\psi_{n\ell m}(r,\theta,\phi) = R_{n\ell}(r)\, Y_\ell^m(\theta,\phi),$$

the probability of finding the electron within radius a_0 is

$$P(r < a_0) = \int_{r<a_0} |\psi|^2\, d^3r = \int_0^{a_0} r^2 |R_{n\ell}(r)|^2\, dr \int |Y_\ell^m|^2\, d\Omega.$$

Because the angular eigenfunctions are normalized on the unit sphere,

$$\int |Y_\ell^m|^2\, d\Omega = 1,$$

then radial-only result is

$$\boxed{P_{n\ell}(r < a_0) = \int_0^{a_0} r^2 |R_{n\ell}(r)|^2\, dr}$$

$2s$ state: $(n,\ell) = (2,0)$

From Table 8.1,

$$R_{20}(r) = 2\left(\frac{Z}{2a_0}\right)^{3/2}\left(1 - \frac{Zr}{2a_0}\right) e^{-Zr/2a_0}$$

For hydrogen ($Z = 1$),

$$R_{20}(r) = 2\left(\frac{1}{2a_0}\right)^{3/2}\left(1 - \frac{r}{2a_0}\right) e^{-r/2a_0}$$

Then

$$P_{2s}(r < a_0) = \int_0^{a_0} r^2 |R_{20}(r)|^2\, dr$$

Let $\rho = r/a_0$ ($0 \le \rho \le 1$), so $r = a_0\rho$ and $dr = a_0\, d\rho$. Also,

$$|R_{20}(r)|^2 = 4\left(\frac{1}{2a_0}\right)^3\left(1 - \frac{\rho}{2}\right)^2 e^{-\rho} = \frac{1}{2a_0^3}\left(1 - \frac{\rho}{2}\right)^2 e^{-\rho}$$

Therefore

$$P_{2s}(r < a_0) = \int_0^1 \left(a_0^2\rho^2\right) \left[\frac{1}{2a_0^3}\left(1 - \frac{\rho}{2}\right)^2 e^{-\rho}\right] (a_0\, d\rho) = \frac{1}{2}\int_0^1 \rho^2 \left(1 - \frac{\rho}{2}\right)^2 e^{-\rho}\, d\rho$$

Evaluating the integral,

$$\boxed{P_{2s}(r < a_0) = 1 - \frac{21}{8e} \approx 3.43 \times 10^{-2}}$$

$2p$ states: $(n, \ell) = (2, 1)$ with $m = -1, 0, 1$

From Table 8.1,

$$R_{21}(r) = \frac{1}{\sqrt{3}}\left(\frac{Z}{2a_0}\right)^{3/2}\left(\frac{Zr}{a_0}\right) e^{-Zr/2a_0}$$

For hydrogen ($Z = 1$),

$$R_{21}(r) = \frac{1}{\sqrt{3}}\left(\frac{1}{2a_0}\right)^{3/2}\left(\frac{r}{a_0}\right) e^{-r/2a_0}$$

Then

$$P_{2p}(r < a_0) = \int_0^{a_0} r^2 |R_{21}(r)|^2\, dr$$

With $\rho = r/a_0$,

$$|R_{21}(r)|^2 = \frac{1}{3}\left(\frac{1}{2a_0}\right)^3 \rho^2 e^{-\rho} = \frac{1}{24a_0^3}\rho^2 e^{-\rho}$$

Thus

$$P_{2p}(r < a_0) = \int_0^1 \left(a_0^2\rho^2\right)\left[\frac{1}{24a_0^3}\rho^2 e^{-\rho}\right](a_0\, d\rho) = \frac{1}{24}\int_0^1 \rho^4 e^{-\rho}\, d\rho$$

Evaluating the integral,

$$\boxed{P_{2p}(r < a_0) = 1 - \frac{65}{24e} \approx 3.66 \times 10^{-3}}$$

This value applies to each of the three m-states ($m = -1, 0, 1$) because m only affects the angular factor, and its contribution integrates to 1 over $d\Omega$.

Comparison ($\ell = 0$ vs. $\ell = 1$)

Numerically,

$$P_{2s}(r < a_0) \approx 3.4\%, \qquad P_{2p}(r < a_0) \approx 0.37\%,$$

so the $2s$ state has about an order-of-magnitude larger probability to be found within one Bohr radius than the $2p$ states.

Problem 2: Emission lines and hydrogenic atoms

Solution

(a) Wavelengths for hydrogen emission lines ($Z = 1$)

For a transition ($n_i \to n_f$) in hydrogen, the emitted photon satisfies

$$\Delta E = E_{n_f} - E_{n_i} = \frac{hc}{\lambda},$$

and with the Rydberg formula (for $Z = 1$)

$$\boxed{\frac{1}{\lambda} = R_\infty \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)} \qquad (n_i > n_f),$$

where $R_\infty \approx 1.097373 \times 10^7 \text{ m}^{-1}$.

(i) **Lyman-α** ($2 \to 1$):

$$\frac{1}{\lambda} = R_\infty \left(1 - \frac{1}{4} \right) = \frac{3}{4} R_\infty \;\Rightarrow\; \lambda = \frac{4}{3R_\infty}$$

Numerically,

$$\boxed{\lambda_{\text{Ly}\alpha} \approx 1.216 \times 10^{-7} \text{ m} = 121.6 \text{ nm}}$$

(ii) **Balmer-α (Hα)** ($3 \to 2$):

$$\frac{1}{\lambda} = R_\infty \left(\frac{1}{2^2} - \frac{1}{3^2} \right) = R_\infty \left(\frac{1}{4} - \frac{1}{9} \right) = R_\infty \frac{5}{36} \;\Rightarrow\; \lambda = \frac{36}{5R_\infty}$$

Numerically,

$$\boxed{\lambda_{\text{H}\alpha} \approx 6.563 \times 10^{-7} \text{ m} = 656.3 \text{ nm}}$$

(iii) **Balmer-β (Hβ)** ($4 \to 2$):

$$\frac{1}{\lambda} = R_\infty \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = R_\infty \left(\frac{1}{4} - \frac{1}{16} \right) = R_\infty \frac{3}{16} \;\Rightarrow\; \lambda = \frac{16}{3R_\infty}$$

Numerically,

$$\boxed{\lambda_{\text{H}\beta} \approx 4.861 \times 10^{-7} \text{ m} = 486.1 \text{ nm}}$$

Problem 2(b): Hydrogenic atoms

A hydrogenic atom consists of a single electron bound to a nucleus of charge $+Ze$. The Coulomb potential is therefore

$$V(r) = -\frac{Ze^2}{4\pi\varepsilon_0 r}$$

The Schrödinger equation with this potential is solved in Chapter 8. Only the dependence on the nuclear charge Z is summarized here.

Energy levels $E_n(Z)$

From the solution of the radial equation, the allowed bound-state energies are

$$E_n = -\frac{1}{2n^2}\left(\frac{Ze^2}{4\pi\varepsilon_0}\right)^2 \frac{\mu}{\hbar^2}, \qquad n = 1, 2, 3, \ldots$$

where μ is the reduced mass of the electron–nucleus system. Compared to hydrogen ($Z = 1$), the energy levels scale as Z^2.

Bohr radius $a(Z)$

In the derivation of the radial equation, a characteristic length scale appears:

$$a = \frac{4\pi\varepsilon_0\hbar^2}{\mu Z e^2}$$

For hydrogen ($Z = 1$), this reduces to the Bohr radius

$$a_0 = \frac{4\pi\varepsilon_0\hbar^2}{\mu e^2}$$

Therefore, for a hydrogenic atom,

$$a(Z) = \frac{a_0}{Z}$$

Rydberg constant $R_\infty(Z)$

Spectral lines are produced by transitions between energy levels:

$$\Delta E = E_{n_f} - E_{n_i} = \frac{hc}{\lambda}$$

Using the expression for $E_n(Z)$,

$$\frac{1}{\lambda} = \frac{\mu Z^2 e^4}{2(4\pi\varepsilon_0)^2\hbar^2 hc}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

This has the standard Rydberg form,

$$\frac{1}{\lambda} = R_\infty(Z)\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

with

$$R_\infty(Z) = Z^2 R_\infty, \qquad R_\infty = \frac{\mu e^4}{2(4\pi\varepsilon_0)^2\hbar^2 hc}$$

Summary

For a hydrogenic atom,

$$a(Z) = \frac{a_0}{Z}, \qquad E_n(Z) \propto -\frac{Z^2}{n^2}, \qquad R_\infty(Z) = Z^2 R_\infty$$

(c) Hα wavelength for $Z = 2$ and $Z = 3$, and spectrum region

Hα corresponds to the transition $(3 \to 2)$, so for charge Z,

$$\frac{1}{\lambda(Z)} = R_\infty Z^2 \left(\frac{1}{4} - \frac{1}{9}\right) = R_\infty Z^2 \frac{5}{36} \;\Rightarrow\; \lambda(Z) = \frac{36}{5R_\infty Z^2}$$

Thus the wavelength scales as

$$\lambda(Z) = \frac{\lambda(Z=1)}{Z^2}$$

Using $\lambda_{\mathrm{H}\alpha}(Z=1) \approx 656.3$ nm:

$$\boxed{\lambda_{\mathrm{H}\alpha}(Z=2) = \frac{656.3}{4}\ \mathrm{nm} \approx 164.1\ \mathrm{nm}}$$

$$\boxed{\lambda_{\mathrm{H}\alpha}(Z=3) = \frac{656.3}{9}\ \mathrm{nm} \approx 72.9\ \mathrm{nm}}$$

Classification:

- 164 nm is **ultraviolet** (far-UV).
- 73 nm is **extreme ultraviolet** (EUV).

Problem 3: Hydrogen wavefunction (Ch. 8.5)

Solution

The given state is

$$\tilde{\psi}(r,\theta,\phi) = \frac{A}{a_0^{3/2}}\left(\frac{1}{\sqrt{\pi}}e^{-r/a_0} - \frac{1}{\sqrt{8\pi}}\frac{r}{a_0}e^{-r/2a_0}\cos\theta\right)$$

(a) Expand in the $\psi_{n\ell m}(r,\theta,\phi) = R_{n\ell}(r)Y_\ell^m(\theta,\phi)$ basis and find A

Using the tabulated full hydrogen wavefunctions:

$$\psi_{100}(r,\theta,\phi) = \frac{1}{\sqrt{\pi}a_0^{3/2}}e^{-r/a_0}$$

$$\psi_{210}(r,\theta,\phi) = \frac{1}{4\sqrt{2\pi}\,a_0^{3/2}}\left(\frac{r}{a_0}\right)e^{-r/2a_0}\cos\theta$$

Compare coefficients:

$$\frac{A}{a_0^{3/2}}\frac{1}{\sqrt{\pi}}e^{-r/a_0} = A\,\psi_{100}$$

and since

$$\frac{1}{\sqrt{8\pi}} = \frac{1}{2\sqrt{2\pi}} = 2\left(\frac{1}{4\sqrt{2\pi}}\right)$$

$$\frac{A}{a_0^{3/2}}\frac{1}{\sqrt{8\pi}}\left(\frac{r}{a_0}\right)e^{-r/2a_0}\cos\theta = 2A\,\psi_{210}$$

Therefore

$$\boxed{\tilde{\psi} = A\left(\psi_{100} - 2\psi_{210}\right)}$$

Normalize:

$$1 = \langle\tilde{\psi}|\tilde{\psi}\rangle = |A|^2\left(\langle\psi_{100}|\psi_{100}\rangle + 4\langle\psi_{210}|\psi_{210}\rangle\right) = |A|^2(1+4)$$

so

$$\boxed{A = \frac{1}{\sqrt{5}}} \quad \text{(taking } A \text{ real and positive)}$$

(b) Time-dependent wavefunction

Hydrogen energy eigenvalues depend only on n:

$$E_n = -\frac{13.6\text{ eV}}{n^2}$$

Here $n = 1$ for ψ_{100} and $n = 2$ for ψ_{210}. Thus

$$\boxed{\tilde{\psi}(r,\theta,\phi,t) = \frac{1}{\sqrt{5}}\left(e^{-iE_1t/\hbar}\psi_{100}(r,\theta,\phi) - 2e^{-iE_2t/\hbar}\psi_{210}(r,\theta,\phi)\right)}$$

(c) Measurements at $t = 0$

(i) Energy

Possible results: E_1 and E_2.

Probabilities:

$$P(E_1) = \left|\frac{1}{\sqrt{5}}\right|^2 = \frac{1}{5}, \qquad P(E_2) = \left|\frac{-2}{\sqrt{5}}\right|^2 = \frac{4}{5}$$

Post-measurement state:

$$E_1 : \ \psi_{100}, \qquad E_2 : \ \psi_{210}$$

(ii) L^2

For ψ_{100}, $\ell = 0$ so $L^2 = 0$. For ψ_{210}, $\ell = 1$ so $L^2 = \ell(\ell+1)\hbar^2 = 2\hbar^2$.

Possible results: 0 and $2\hbar^2$

Probabilities:

$$P(L^2 = 0) = \frac{1}{5}, \qquad P(L^2 = 2\hbar^2) = \frac{4}{5}$$

Post-measurement state:

$$L^2 = 0 : \ \psi_{100}, \qquad L^2 = 2\hbar^2 : \ \psi_{210}$$

(iii) L_z

Both terms have $m = 0$, so the only possible result is

$$\boxed{L_z = 0} \quad \text{with} \quad P(L_z = 0) = 1$$

Post-measurement state:

$$\tilde{\psi} = \frac{1}{\sqrt{5}}\left(\psi_{100} - 2\psi_{210}\right)$$

Time-varying probability density

After an energy measurement, the state is ψ_{100} or ψ_{210} (stationary), so $|\psi|^2$ does not depend on t.

After an L^2 measurement, the state is also ψ_{100} or ψ_{210}, so $|\psi|^2$ does not depend on t.

After an L_z measurement, the state remains a superposition of different n values, so the relative phase changes in time and $|\tilde{\psi}(r, \theta, \phi, t)|^2$ is time-dependent.

$$\boxed{|\psi|^2 \text{ is time-varying after the } L_z \text{ measurement (and in the original state).}}$$

Problem 4: McIntyre 8.12 (hydrogenic systems)

Solution

Use the hydrogenic formulas (nonrelativistic Coulomb model) with reduced mass μ:

$$\mu = \frac{m_1 m_2}{m_1 + m_2}$$

Ground-state energy:

$$E_1(Z) = -13.6 \text{ eV} \left(\frac{\mu}{m_e}\right) Z^2$$

Effective Bohr radius from Eq. (8.8):

$$a(Z) = \frac{4\pi\varepsilon_0\hbar^2}{\mu Z e^2} = \left(\frac{m_e}{\mu}\right)\frac{a_0}{Z}, \qquad a_0 = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2}$$

Lyman-α is the $n = 2 \to 1$ transition, so $\Delta E = \frac{3}{4}|E_1|$ and

$$\lambda_{\text{Ly}\alpha}(Z) = \frac{hc}{\Delta E} = \frac{121.6 \text{ nm}}{\left(\frac{\mu}{m_e}\right) Z^2}$$

Mass ratios used (in units of m_e):

$$m_p \simeq 1836.15\, m_e, \quad m_d \simeq 3670.48\, m_e, \quad m_\alpha \simeq 7294.30\, m_e, \quad m_\mu \simeq 206.77\, m_e$$

(a) Deuterium: $e^- + \text{deuteron}$ $(Z = 1)$

$$\frac{\mu}{m_e} = \frac{m_d}{m_d + m_e} \approx 0.999728$$

$$E_1 \approx -13.596 \text{ eV}, \qquad a \approx 1.00027\, a_0, \qquad \lambda_{\text{Ly}\alpha} \approx 121.60 \text{ nm}$$

(b) Positive helium ion: $^4\text{He}^+$ $(Z = 2)$

$$\frac{\mu}{m_e} = \frac{m_\alpha}{m_\alpha + m_e} \approx 0.999863$$

$$E_1 \approx -54.393 \text{ eV}, \qquad a \approx 0.50007\, a_0, \qquad \lambda_{\text{Ly}\alpha} \approx 30.40 \text{ nm}$$

(c) Positronium: $e^- + e^+$ $(Z = 1)$

$$\mu = \frac{m_e m_e}{m_e + m_e} = \frac{m_e}{2} \quad \Rightarrow \quad \frac{\mu}{m_e} = \frac{1}{2}$$

$$E_1 = -6.80 \text{ eV}, \qquad a = 2a_0, \qquad \lambda_{\text{Ly}\alpha} \approx 243.13 \text{ nm}$$

(d) Muonium: $e^- + \mu^+$ $(Z = 1,\ m_\mu = 207\, m_e)$

$$\frac{\mu}{m_e} = \frac{m_\mu}{m_\mu + m_e} \approx \frac{206.77}{207.77} \approx 0.995187$$

$$E_1 \approx -13.535 \text{ eV}, \qquad a \approx 1.00484\, a_0, \qquad \lambda_{\text{Ly}\alpha} \approx 122.15 \text{ nm}$$

(e) Muonic hydrogen: $\mu^- + p$ ($Z = 1$)

$$\frac{\mu}{m_e} = \frac{m_\mu m_p}{m_\mu + m_p}\frac{1}{m_e} = \frac{(206.77)(1836.15)}{206.77 + 1836.15} \approx 185.841$$

$$E_1 \approx -2.53 \times 10^3 \text{ eV}, \qquad a \approx 0.00538\, a_0, \qquad \lambda_{\text{Ly}\alpha} \approx 0.654 \text{ nm}$$

(f) Hydrogen-like uranium: $^{235}\text{U}^{91+}$ (one electron, $Z = 92$)

Here $Z = 92$ (uranium nuclear charge). The nucleus is very heavy so $\mu/m_e \approx 1$.

$$E_1 \approx -13.6 \text{ eV}\,(92)^2 \approx -1.15 \times 10^5 \text{ eV}$$

$$a \approx \frac{a_0}{92} \approx 0.01087\, a_0, \qquad \lambda_{\text{Ly}\alpha} \approx \frac{121.6 \text{ nm}}{92^2} \approx 0.0144 \text{ nm}$$

Problem 5: McIntyre 8.14 (superposition state)

Solution

The initial state is

$$|\psi(0)\rangle = \frac{1}{\sqrt{14}}\,|2,1,1\rangle - \frac{2}{\sqrt{14}}\,|3,2,-1\rangle + \frac{3i}{\sqrt{14}}\,|4,2,2\rangle\,.$$

The basis kets $|n,\ell,m\rangle$ are orthonormal, so measurement probabilities come from the squared magnitudes of the expansion coefficients:

$$\left|\frac{1}{\sqrt{14}}\right|^2 = \frac{1}{14}, \qquad \left|\frac{-2}{\sqrt{14}}\right|^2 = \frac{4}{14}, \qquad \left|\frac{3i}{\sqrt{14}}\right|^2 = \frac{9}{14}.$$

(a) Energy: outcomes, probabilities, histogram, and $\langle E\rangle$

For hydrogen, the energy depends only on n:

$$E_n = -\frac{13.6\text{ eV}}{n^2} \qquad \text{(hydrogen)}.$$

Thus the possible energy outcomes are E_2, E_3, E_4 with probabilities

$$P(E_2) = \frac{1}{14}, \qquad P(E_3) = \frac{4}{14}, \qquad P(E_4) = \frac{9}{14}.$$

Expectation value:

$$\langle E\rangle = \frac{1}{14}E_2 + \frac{4}{14}E_3 + \frac{9}{14}E_4.$$

Using $E_2 = -3.4$ eV, $E_3 = -13.6/9$ eV, $E_4 = -13.6/16$ eV,

$$\boxed{\langle E\rangle \approx -1.221\text{ eV}.}$$

Histogram (TikZ):

[ybar, bar width=18pt, xlabel=n (energy level), ylabel=Probability, xtick=2,3,4, ymin=0, ymax=0.75, width=0.65height=0.35] coordinates (2,1/14) (3,4/14) (4,9/14);

Figure 1: Energy measurement probabilities.

(b) L^2: outcomes, probabilities, histogram, and $\langle L^2\rangle$

For $|n,\ell,m\rangle$,

$$\hat{L}^2\,|n,\ell,m\rangle = \ell(\ell+1)\hbar^2\,|n,\ell,m\rangle\,.$$

The three components have $\ell = 1$ (first term) and $\ell = 2$ (second and third terms). So

$$L^2 = 2\hbar^2 \;\; (\ell = 1), \qquad L^2 = 6\hbar^2 \;\; (\ell = 2).$$

Probabilities:

$$P(L^2 = 2\hbar^2) = \frac{1}{14}, \qquad P(L^2 = 6\hbar^2) = \frac{4}{14} + \frac{9}{14} = \frac{13}{14}.$$

Expectation value:

$$\langle L^2 \rangle = \frac{1}{14}(2\hbar^2) + \frac{13}{14}(6\hbar^2) = \frac{40}{7}\hbar^2.$$

$$\boxed{\langle L^2 \rangle = \frac{40}{7}\hbar^2.}$$

Histogram (TikZ):

[ybar, bar width=26pt, xlabel=L^2 outcome, ylabel=Probability, symbolic x coords=$2\hbar^2$,$6\hbar^2$, xtick=data, ymin=0, ymax=1.0, width=0.65height=0.35] coordinates ($2\hbar^2$,1/14) ($6\hbar^2$,13/14);

Figure 2: L^2 measurement probabilities.

(c) L_z: outcomes, probabilities, histogram, and $\langle L_z \rangle$

For $|n, \ell, m\rangle$,

$$\hat{L}_z |n, \ell, m\rangle = m\hbar |n, \ell, m\rangle .$$

Here the three components have $m = 1, -1, 2$. So the possible outcomes are

$$L_z = \hbar, \quad L_z = -\hbar, \quad L_z = 2\hbar$$

with probabilities

$$P(L_z = \hbar) = \frac{1}{14}, \qquad P(L_z = -\hbar) = \frac{4}{14}, \qquad P(L_z = 2\hbar) = \frac{9}{14}.$$

Expectation value:

$$\langle L_z \rangle = \hbar \left[\frac{1}{14}(1) + \frac{4}{14}(-1) + \frac{9}{14}(2) \right] = \hbar \left(\frac{15}{14} \right).$$

$$\boxed{\langle L_z \rangle = \frac{15}{14}\hbar.}$$

Histogram (TikZ):

[ybar, bar width=18pt, xlabel=L_z outcome, ylabel=Probability, symbolic x coords=$-\hbar$,$\hbar$,$2\hbar$, xtick=data, ymin=0, ymax=0.75, width=0.65height=0.35] coordinates ($-\hbar$,4/14) ($\hbar$,1/14) ($2\hbar$,9/14);

Figure 3: L_z measurement probabilities.

(d) Time dependence of (a), (b), and (c)

Time evolution adds phases to each energy eigenstate:

$$|\psi(t)\rangle = \frac{1}{\sqrt{14}} e^{-iE_2 t/\hbar} |2, 1, 1\rangle - \frac{2}{\sqrt{14}} e^{-iE_3 t/\hbar} |3, 2, -1\rangle + \frac{3i}{\sqrt{14}} e^{-iE_4 t/\hbar} |4, 2, 2\rangle .$$

The magnitudes of the coefficients do not change with time, so the probabilities in (a), (b), and (c) are time-independent. The expectation values $\langle E \rangle$, $\langle L^2 \rangle$, and $\langle L_z \rangle$ are also time-independent.